The possibility to use gift as a paradigm for exchanges among agents

J. Rouchier, F. Bousquet[2]

Centre for Policy Modelling - Manchester Metropolitan University
http://www.cpm.mmu.ac.uk/~juliette/
j.rouchier@mmu.ac.uk

Abstract. In the paper, we describe artificial societies in which interactions among agents are gifts. The dynamics described is inspired by examples of real societies. Gifts exist as a system that is distinct from merchant exchange. The main principle is that goods have no importance for people involved and that only the act of giving has a value. Hence in these systems, there exists an evaluation of each one by the others (reputation), based on its participation to the giving dynamics. One distinguishes between two types of gifts: one to sustain those that are less prestigious, another one to get prestige.
Following these ideas, we built a gift system in a society of artificial agents, each being defined by its rank (ranking of reputations), its motivation to get prestige and its esteem (ability to give). We made simulation to study the orders that emerge in these societies, and to see how one could reinforce them by having the choices of agents evolve depending on the social position. Some very stable patterns could be identified, showing to what extent the rules described might be used to organize a system where reputation would matter and could be considered as a social construction and not just an individual characteristic.

1 Introduction

The aim of the paper is to describe the main interaction processes on which gift economy is grounded. We show how stable organisations can emerge in an artificial society in which agents would perform these gifts and how one can relate the appearing order to the value of some key parameters. One sees a relation between assumptions on agents' goals and the patterns of society and this enables us to understand how the emerging regularity could be use to produce architectures for multi-agent system.

Ethnologists are interested in the study of gifts exchange that they consider as a good tools for the reproduction of habits, values and structures in a society [8]. The first consequence of a gift is to create a link between the people involved.

---

[2] Cirad-Tera-Ere, Montpellier, France, bousquet; lepage @cirad.fr

The main logic that enables a system of that kind to work is that individuals are not necessarily interested in possessing a lot of values. The ability to give away is indeed as important as the ability to accumulate goods. One can see it in the very extreme behaviours chiefs in some societies challenging each other by doing gifts fights to accumulate prestige (the more one gives, the more prestigious) [2]. A more normal behaviour shows that being able to give away is socially positive: the role of rich members of families or groups is to help the others by redistributing their goods. To be generous is a social quality that is recognised by all [11].

In some cultures, a gift has to be given back. By performing a gift, one shows that one is willing to engage in a relation and believes that the other will understand the message and express this understanding by giving back. The reciprocation creates a never-ending loop of encounters that establishes a stable relation. In some societies, one has to give even more than one received and the relation can then turn out to be really competitive. We decided here not to focus on the reciprocal characteristics of the gift, which had been studied in a previous work [14, 15]. In that former study we analysed interdependences in a group due to gifts and counter-gifts, observing the hierarchies that can be induced by inter-individual links. It is also to note that there exists quite a lot of cases where giving back is not necessary[1].

Most examples of the gift exchanges come from foreign societies [2, 8, 10, 11] since they are the one that were studied by ethnographers. Actually, one can note that the gift (and ostentatious behaviours in general) is also very important in our cultures and contributes to the image of each individual when the others consider him or her. Indeed, the main idea put forward by the ethnologists is that exchanges of goods create symbolic values in a group. Based on that value that is different from material possession, although not totally independent, there exist different kind of social order. This is an idea that we wanted to develop and support by experiments led in artificial societies.

The work presented here shows what kind of stability of images can appear in the context of an exchange system. We choose to represent gift system for two reasons: it is very important in societies and has an impact on social order which is clear and well documented; it is different from the paradigms used in MAS systems, where interactions are mainly influenced by merchant exchange. While building the system we also wanted to see if individual agents could be characterised individually, although all dynamics were purely social. Simulation indeed seems to be a good way to understand how models of society can lead to certain global structures that have then an retro-action on the individual [3, 4, 7, 8, 12, 13].

[1] In a paper in which he discusses different motivations for gift, van de Ven [16] explores different cases when gifts are not given back.

2 Model and implementation

2.1 Conceptual model and assumptions

In the present work, we describe how reputation (as a common representation of each other which is shared among agents) can be used to structure a society in a stable way, as well as influence agents' goals and self-image. For that we built a model of society and organised simulations by changing different parameters and observing the results.

In the system, agents don't worry about who they make the gift to, or whether it will get one in return. The gift is seen as compulsory and agents just try to conform to that[2]. That is why the only relevant information for the agent is to know about others' actions. Another consequence is that they give away all the goods that they don't consume, doing one gift at each time-step. The reputation is based on the number of gifts made and received. It depends of the type of gift made as well.

Indeed, there exist two types of gifts that create two types of reputation: the prestige gift and sharing gift (to show one is well integrated). One does a prestige gift to get prestige. Conversely, agents make prestige gifts to the one they consider as prestigious, therefore receiving a prestige gift indicates that one is recognised as having prestige. If I share with someone, it is because I want to show that I want to be integrated, but also that I consider the other as integrated in the group. Both giving and receiving gifts improves reputation.

An agent is characterised individually by its reputation: prestige reputation and integration reputation, and by the two parameters that make it choose its actions: the motivation (make gifts either to get prestige or to share), and its esteem (belief in the ability to perform gifts that are acceptable by the group). The agents see those two criteria evolve due to their former actions and their reputation in the group. Therefore individual history is defined by the evolution of reputation, motivation and esteem.

All the agents do not have the same interest in prestige and integration. Hence some are more motivated to share with the others whereas others look for prestige. Here, the motivation was considered as dependent of the position in the group in a very direct way that tends to reinforce the actual situation. It seems unlikely that a very badly integrated agent should be often allowed to make a gift to the most prestigious ones. A priori, the more integrated, the more an agent may participate to social dynamics: this is why the esteem was linked to the reception and emission of gifts.

[2] In real societies, when gift is important, the value of an individual in the society is not related to his possessions but to his ability to give away [17], and this is why it was chosen here that only the participation to the common dynamics grounds the evaluation of the agents.

In that study, we wanted to stress the circularity between the choices made individually and the social situation in which the agent is. The hierarchy reveals the recent history of exchanges that took place in the society; conversely each act is strictly influenced by the social order. The multi-agent system was established in order to perform simulations that would enable the apparition of that circular process[3].

When choices are fixed for each agent, it is possible to build a reputation that does make sense: one can for example observe societies with stable distinct groups, like elites constituted of agents that have clearly a higher reputation of prestige than the others. A link was made between reputation and motivation for prestige and between a local perception of success and failure and the esteem. This double loop seems to give an even greater coherence to the society whereas some surprisingly complex patterns can still be observed.

2.2 Agent and definition of a time-step

The society consists of a group with 50 agents, each characterised by its: prestige reputation ; integration reputation ; rank (rank 1 is the agent with the highest reputation of prestige) ; motivation for prestige (between 0 and 10) ; motivation for sharing (motivation for prestige - 10) ; esteem (between 0 and 10).

All agents consume the same minimal amount of money at each time step. They earn money by working (which gives twice the minimum value for consuming) and/or by receiving gifts at the previous time-step (since they spend all their money and then receive gifts). For a sharing gift, the agent shares what it possesses in 2 equal parts and keeps half for its own consumption. For a prestige gift, the agent shares its money in three, two third being for the gift. An agent may do a prestige gift only if it already has enough money: if it has to work it will have to turn to a sharing gift. Agents choose the gift they want to perform depending on their choice parameters (motivation for prestige and esteem) (Figure 1). The group designates the receptor: a sharing gift is given randomly to an agent with a rank which value is more than (rank of the giving agent - 2): they are agents with less prestige or equivalent; a prestige gift is given randomly to an agent which rank is less than (rank of the giving agent + 2).

Once the gifts have been made, the new reputation of each agent is calculated locally: the prestige reputation (integration) is the sum over 25 time steps of all reputation (integration) tokens that are accumulated at each time step by an agent (Figure 2). Then the group establishes the ranks by comparing the reputations: the first rank has the highest prestige reputation, and the last rank has the lowest. The memory of the group and of the agents was chosen to be 25 steps for all the simulations made.

The choice parameters evolve along the time, since we want the agent's rationality to depend on its social position. They thus evolve depending on the acts of the time step and on the reputation:

Motivation for prestige / motivation for sharing =
(motivation change constant) * (prestige reputation / integration reputation)

---

[3] We implemented the model in Smalltalk, using Visual Work 3.0.

The evolution of the esteem is such that if the agent makes any kind of gift, its esteem increases by 0.1; if the agent receives no gift its esteem decreases by 0.2.

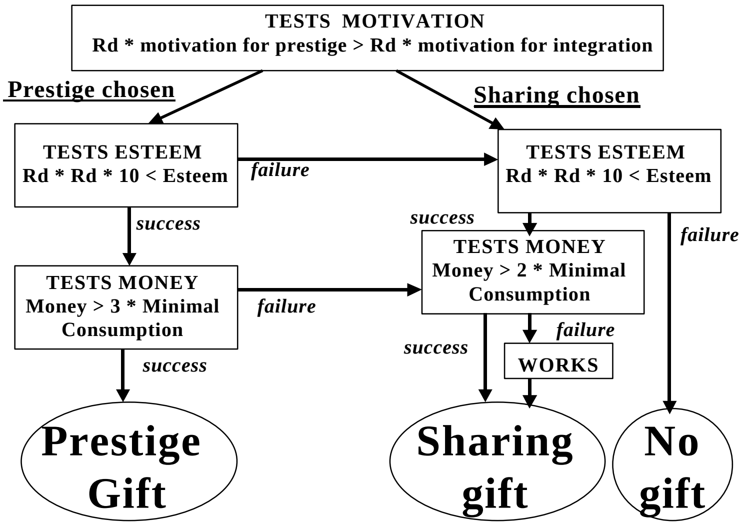

Fig. 1. How the agent tests all its attributes to decide which gift it is eventually going to do. "Rd" means a random number between 0 and 1. The two motivations are always such that: motivation for prestige + motivation for sharing = 10. Each agent calculates it according to its characteristics.

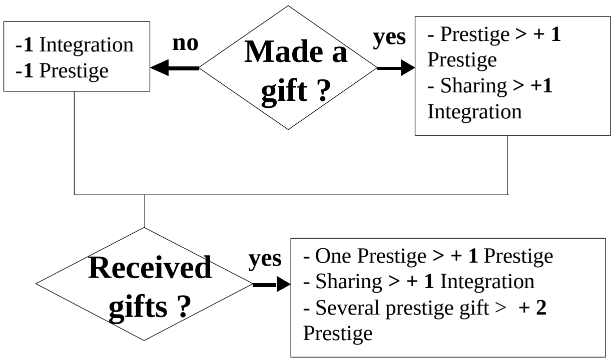

Fig. 2: How the group calculates the increase or decrease of reputation for each agent before calculating ranks

2.3 Simulations and definition of observation

Simulations were successions of 1000 steps. They were defined as follow:

Table 1: the different types of simulations.

Type of simulation	**Motivation for prestige**	**Esteem**
Basic[4]	The same for all agents, motivation for prestige between 0 and 10	The same for all agents, value between 2 and 8
Motivation evolves	Motivation evolves with reputation	The same for all agents, value between 2 and 8
Esteem evolves	The same for all agents, motivation for prestige between 0 and 10	Esteem evolves with gifts made and received

The aim while building the system was to be able to build reputation and to see if it could characterise one individual agent itself. We thus observed differences between the agents in terms of reputation, and the regularity of the ranking. We were also interested in studying globally the number of gifts and their type for each turn, to be able to characterise a society with a lot of exchanges or, on the contrary, societies with little circulation of goods. The criteria used could thus be individual or global.

Individual criteria: Evolution of both reputations and rank, of the motivation, of the esteem, Number of gifts received in 25 steps, Number of gifts made in 25 steps

Global criteria: Number of agents that possess each given value of reputation, integration reputation as a function of prestige reputation (this helps us to identify agents that would have a very different rank from the other at a given time step), number of gifts of each category at each time step.

The elements were observed for 1000 steps. For each case, a direct observation was performed for all the criteria, and then some of simulations were repeated 30 or 50 times. The results were almost always similar for the same initial setting, except for some cases that are indicated.

2.4 Patterns of reputation repartition: types of societies and differentiation

There exist three main types of societies that appear depending on the initial conditions, when esteem and motivation are fixed:

- prestige reputation is equivalent and all agents can hardly be distinguished;
- a group is clearly identified, called here an elite, where agents have a prestige reputation that is much higher than the rest of the group;
- some agents have a higher prestige reputation, but the elite is not separated clearly from the rest and agents have no stable rank.

[4] The simulations where esteem and motivation evolve interested us mainly, but it was necessary to create a previous characterisation of possible patterns in the society and relevant indicators. Thus we did a few simulations where the choice parameters were fixed, called here “easic simulations”.

For these simulations, esteem does not structure the repartition of gifts in the group, it is more a limiting variable. A minimum esteem is necessary in the group to have the gifts actually performed according to the agents' will, and therefore see differentiation occur.

There exists simulation outcomes with no real differentiation between agents: if there are not enough exchanges to have differences in reputation, it is a "low participation" society; if agents do not want prestige and most of the gifts made are sharing gifts, one talks of a '"sharing" society; if all agents want to be prestigious and have an esteem that is high enough to have a lot of gifts circulate, no hierarchy ever stabilises and one gets a "challenging" society.

Conversely, one can also observe quite heterogeneous societies. For certain initial conditions one witnesses the apparition of an elite of four agents, more or less separated from the rest of the population. The agents that are in it have a much higher prestige reputation than the rest of the group. With their low ranks (high reputation is low rank), they are those who receive the main part of the prestige gifts, and thus their situation is well established and constantly reaffirmed. The elite does change, though, but the speed of change depends at the same time on the motivation and on the esteem of the whole population. The mobility of the agents going between the elite and the rest of the population is what creates an intermediary group going in both direction in the order: we call it the "passage group".

Stability here is what designates an order that can be observed unchanging for more than 50 steps. The time scale can vary a lot between the different simulations.

Table 2: types of initial settings depending on values of esteem and motivation for prestige. The observation was led running at least 50 simulations in each case.

	Esteem >= 6	**Esteem < 6**
Motivation for prestige <= 5	No elite, prestige reputation and sharing one quite high and variable. A lot of gifts.	Very few gifts, all reputation values are low
6 <= Motivation for prestige <= 9	Well separated elite, quite stable	Elite well separated often changed
Motivation for prestige = 10	Continuity between a group with no reputation to the elite	Small differentiation for an unstable elite

3 Simulations where esteem evolves

In the simulation, populations are homogenous at the beginning, and defined by the motivation of the agents and the initial value of the esteem. Different evolution of esteem can appear, that create different orders in the society. We describe first the global evolution of motivation and then the kind of order can be related to the distribution of motivation among the agents.

If agents have a low motivation for prestige, all converge quickly to the same level of esteem, either very low or very high. It is not possible for the agents to differentiate themselves regarding their esteem. The homogeneity is settled after 200 time steps, even if esteem is not necessarily stable.

The way the feed-back is defined can explain that result: if a majority of agents have a low esteem they do few gifts and a lot of agents will not get any and thus see their esteem decrease, even if it was initially high. Conversely when a lot of gifts circulate, agents always get a few, that are sufficient to have their esteem grow.

Table 4: value of the esteem for a motivation of 4 or 5

Initial esteem	**Esteem after 200 steps in a simulation with motivation for prestige of 5**
< 6	All esteems fall quickly and stay at the lowest level (around £), almost homogenous.
= 6	Esteem oscillates for all agents between 2 and 8 – except for 1 out of 20 simulations when esteem drops for all.
> 6	All agents gain esteem: almost homogenous and more than 7 for all agents

A different situation appears when agents have a high motivation for prestige and the differentiation is easier: a small group has a higher esteem than the rest of the population, some attaining the maximum value.

Table 5: value of the esteem for a motivation of 8

Initial esteem	**Esteem after 1000 steps in a simulation with motivation for prestige of 8**
< 3	Esteem falls: almost homogenous, less than 3
3-4	Esteem increases for a small group, whereas the others have a low esteem
> 4	Esteem increases for all: almost homogenous, more than 7

The higher the motivation for prestige, the more the esteem will globally increase from a low level. It can be deduced directly from the feed-back definition and the way the agent chooses: any time an agent tries to do a prestige gift, if it fails, it has the opportunity to try to do a sharing gift and has thus twice the chance to succeed and hence not loose any esteem.

For two special values of the esteem (3 or 4) it is possible to identify a new phenomenon. All agents do not have the same esteem after a while: some have a higher esteem all along the simulation and their reputation always stays very stable and no other agent, though motivated, succeeds in getting into the elite (Figure 3).

If the esteem falls down for all agents, the society is alike one with very low esteem at the beginning, whatever the motivation for prestige: very few gifts circulate and there is neither differentiation nor stabilisation. If agents have a high esteem and motivation for prestige is low, a lot of gifts circulate and agents' ranks vary a lot. If they have high esteem with high motivation for prestige, an elite does appear, that is usually quite stable. They are anyway more stable when there is reinforcement in the esteem than with the same average value of esteem and motivation with stable value (Figure 4, Figure 5). In the only case where the esteem is not stable, the society exhibits no differentiation in prestige, like for any population with low motivation for prestige; the oscillation in ranks is the highest we have ever witnessed.

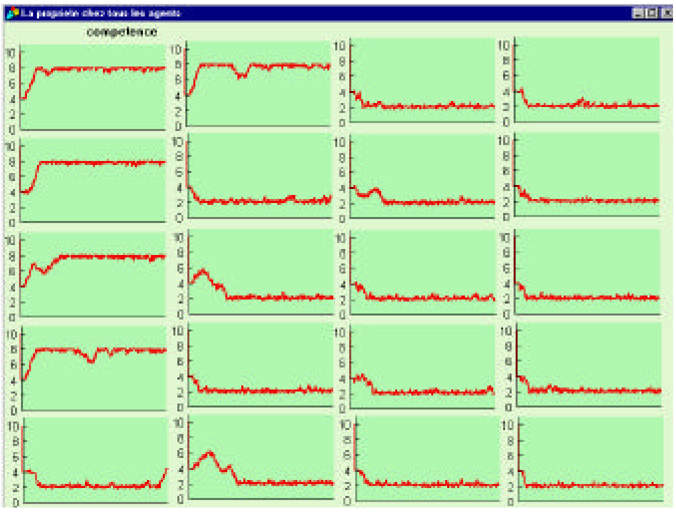

Figure 3: Esteem of 20 highest-ranking agents over 1000 time-step, motivation = 8

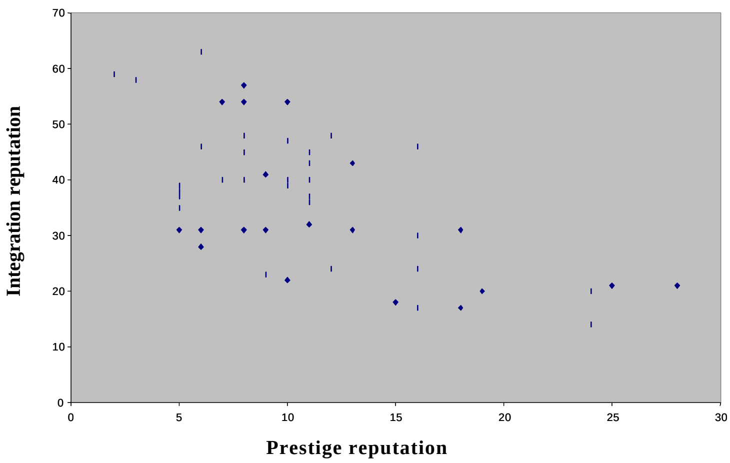

Figure 4: Reputation of prestige and integration for agents that all have a high esteem. An elite can be distinguished with 4 agents (not necessarily the same all the time).

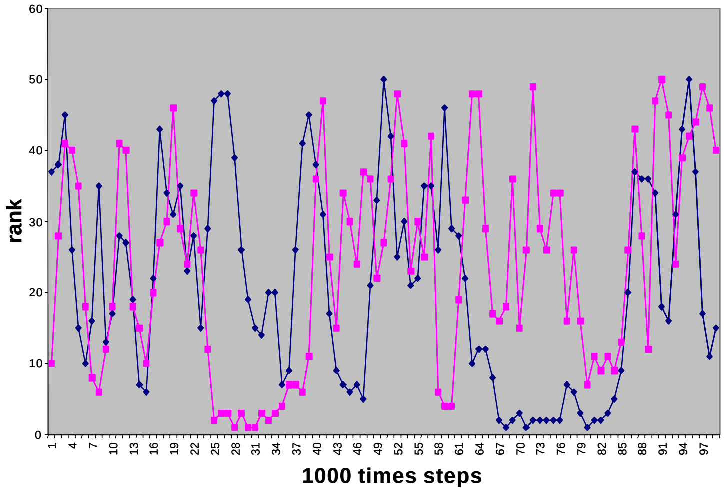

Figure 5: Evolution of the rank of two agents that have evolving esteems beginning at 6 in a society where motivation is 5. No stability: both stay only for a short time in the elite.

For the few simulations in which esteem itself is a differentiating element for agents, the elite is more stable than for the first set of simulations: there are few changes in status, whereas there are a lot if esteem is fixed with equivalent average value. The agents of the elite are easy to identify (Figure 6). One can understand that the elite is here reinforced by the whole group of agents that are all willing to get prestige: the gifts that they make go mainly to the elite agents and thus helps them keeping their esteem (Figure 7).

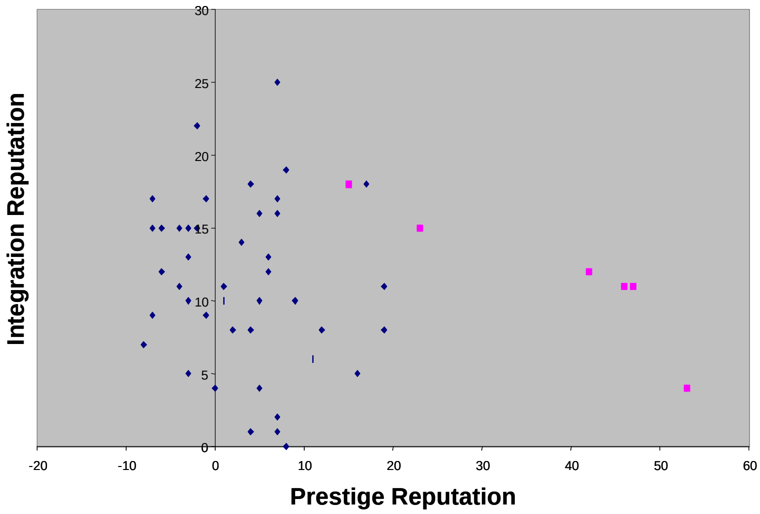

Figure 6: Integration reputation as a function of prestige reputation in a simulation where esteem evolves from 4 and where motivation for prestige is 8. The 6 agents with high esteem

are represented by squares (four or them constitute the elite that is clearly identified and very stable (see table 3).

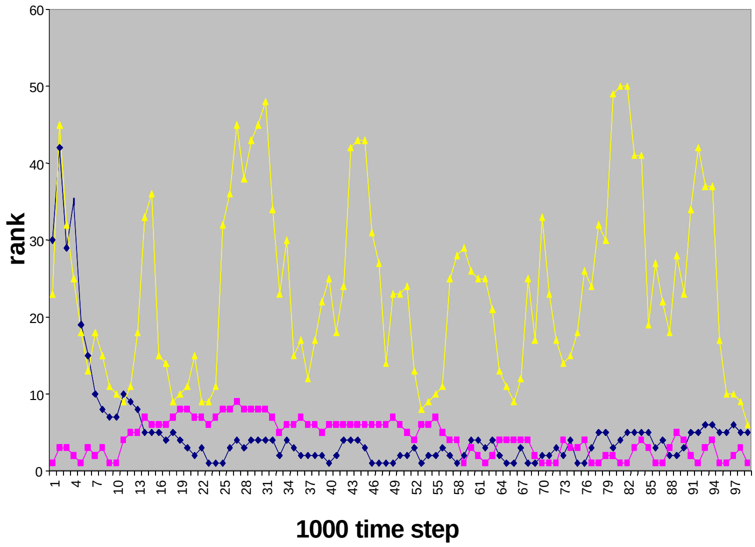

Figure 7: Ranks of three agents during one simulation where esteem varies beginning at 4 with a motivation for prestige of 8. Two of the agents are taken among the 6 ones that have a high esteem (here dark curves): they stay in the elite forever as soon as they got in. The other agent has low esteem all along (clear curve): it cannot stay among the 10 first ranks.

4 Simulations where motivation evolves

In these simulations agents can change motivation at each step, and that motivation is calculated according to the reputation of the previous time-step. We describe first the global evolution of motivation and then the kind of order can be related to the distribution of motivation among the agents.

Each simulation is characterised by two parameters: the motivation constant and the esteem of agents. One identifies several patterns regarding the evolution of motivation for the agents and these patterns then influence the order that appear in the society and thus the dynamic of exchanges (Table 6).

Table 6: Evolution of the motivation when the esteem is 8 (maximal value), depending on the motivation change constant.

Constant	**Evolution of motivation for prestige**
<= 5	Motivation of all agents falls, until the average is 2 after 800 steps. Individually, the motivations can increase and decrease for a while, but eventually none is above 6 after 900th step.

6 - 7	Only 10 agents keep a high motivation for prestige. After 800 simulation steps there are only 10 candidates to prestige, but this group changes: some are replaced and all agents see their motivation oscillate. The average of motivation in the group is less than 3.
>= 8	All motivations for prestige are above 6. For all agents, it is maximal and falls very occasionally. The average is very stable, slightly less than 9.

Table 7: Evolution of the motivation when the esteem is 5, depending on the motivation change constant.

Constant	**Evolution of motivation for prestige**
<= 9	Motivation for prestige falls for all agents. It takes about 1500 steps so that no agent is tempted by prestige anymore.
10 - 13	A minority always has a very high motivation, and those can be replaced every 100 steps. (The situation takes about 1500 steps to be stable. If constant is 10, there are only 5 to 10 agents that want prestige. If it is 13, they are about 20-25.)
>= 14	All agents want prestige almost since the beginning. All motivations fall and go back to maximum.

If the esteem is very low, no agent is ever motivated for prestige for a long time. Even if at one step an agent might be willing to get prestige, it never lasts, it is only a sign that this agent had a very low integration reputation, but it does not influence deeply its actions in a long term.

There are thus patterns of societies that can be related to the way motivation is spread in the group. In these simulations with fixed esteem and evolving motivation, the stabilisation of the motivation takes a very long time and the order takes a longer time to appear than in previous simulations (in former ones, a global and stable structure could be seen after 400 steps, here it needs at least 800). If the esteem is less than 5, the society is totally fixed; without gifts and thus no differentiation (”low participation”). We shall thus be interested in cases where the esteem is more than 5, and then the order depends directly on the evolution of the motivation.

If the motivation for prestige is low for all, we get a “sharing society”.

If a small number of agents are interested in acquiring prestige there is a small elite (about three agents like in Figure 16), which is rather stable. The fewer agents are tempted by prestige, the more stable the elite. If there are more than 15 of those, the elite is very unstable. The society is differentiated into two groups by the number of prestige gifts received. There is the elite and the rest. Some are among the 10 highest ranking agents for a long time. What is quite amazing is that, contrary to the case when a minority has a much higher esteem than the rest of the group, here, there never exists a stable elite to which agents belong for the whole time. Any agent who wants prestige can reach the elite.

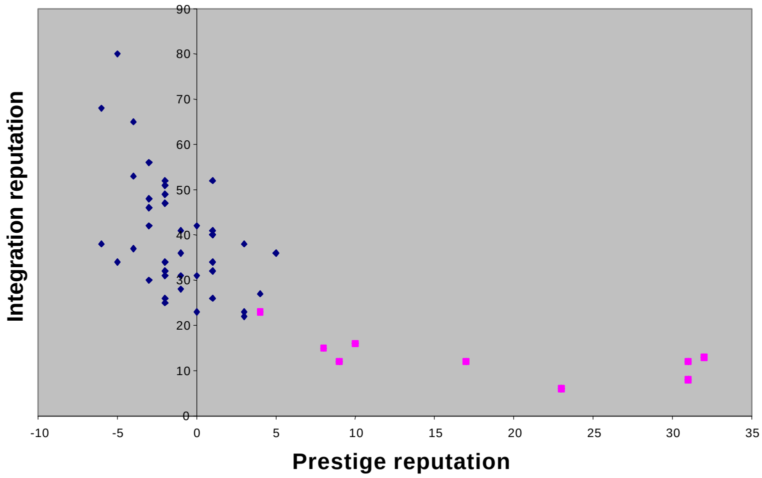

Figure 8: Sharing reputation as a function of prestige reputation in a simulation where esteem is 5 and variable constant is 9. A minority is tempted by the prestige for a long time (clear squares) and constitute a stable elite. In dark: agents that prefer to share.

If the number of agents who want prestige is high, an elite appears, in which agents stay for a long time. The order is similar to a society in which all agents are motivated by prestige and have enough esteem to try to get to the highest rank. Here we can notice a new situation: the minority that do not care about prestige constitute a rather small group, very distinguishable, very stable for long periods. They all have a very low reputation for prestige. On the contrary, the groups of agents that look for power are rarely stable, and their agents change all the time. It is the first time that such a group appears: the feedback on motivation can thus exclude some agents from the dynamics of getting prestige.

5 Discussion

5.1 "personality" of agents and social values

The work presented here is a translation of dynamics observed in real societies in which non-merchant exchanges are important. To enable the evolution of each individual agent, two feedback loops have been defined. The first one transforms the agents' esteem that evolves according to the gifts made and received. The other is motivation for prestige that is calculated with the reputation. Some others loops can be logically deduced from the dynamics of the system:

- Agents with a high prestige reputation get more prestige gifts, which helps keeping their reputation high

- Agents with a high esteem make gift easier, which keeps their esteem high,
- Agents with high reputation for prestige will be motivated by prestige and will act to increase their reputation for prestige.

What we see is that these global loops logically reinforce the dynamics, and we could believe that a lot of deadlocks would thus be created, with two main situations: a high stability of hierarchy, fixed because they repeat themselves, a total randomness, in which it is impossible to identity any pattern. But we could also detect a great range of situations (among which some were not necessarily interesting for our concerns). The main phenomenon is that most of the time motivation for prestige and esteem have dynamics that go together. Both help the circulation of gifts, and then the apparition of reputation of prestige.

Table 8: All situations that are encountered in simulations, and conditions under which they occur (basic = fixed, variable = either esteem or motivation evolves).

	No differentiation	**Two groups: an elite and others**
No clear stability of agents in the elite	Basic: Motivation <= 5 / any esteem- Variable: Motivation <= 5 / Evolving esteem	
Stability for some agents in the first ranks for 20 steps	Variable: Motivation >= 6 / Evolving esteem < 3 at the beginning	Basic: Motivation = 9-10 / Esteem <= 5 - Variable: Motivation >= 9/ Evolving esteem > 7
Stability of some agents in the first ranks for 20 to 200 steps	Basic: motivation = 9-10 / Esteem >= 6 - Variable: Motivation >= 6 / Evolving esteem > 5	Basic: Motivation = 7-8/ Esteem <= 5
Stability of some agents in the first ranks for 500 steps or more		Basic: fixed motivation = 6-8 / esteem >= 7 - Variable: motivation >= 6 / Evolving esteem 3-4 at the beginning

There is a great importance of the parameters of the system, but there are only very few cases where the global results are not predictable from the initial conditions of the simulation. For example, one knows that the order in the society is less stable if the whole population wants to get some prestige and feels able to get it. But cases where one can control the existence of elite and insure predictable ranking are easy to find.

In each society built, an agent is defined by some attributes, and we tend to think as their “personality” the two ones that determine how they act: esteem and motivation. The latter could be seen more or less as the preferences often identified by economists. In a more social way we even identified some “function” for an agent, where its actions take an identified place in the others' reputation evolution in the long term. For example, when some agents cannot be part of the elite, although they want to: they make gifts that help other ones to stay in that elite. In the societies built, if it is possible to identify individual “personalities”, it is impossible to see the reputation of one as an individual result, without considering the whole group.

If esteem evolves, it is almost impossible that one agent could have a high esteem in the long term when others all have low esteem: at least a few of the others must be self-confident as well, there is a minimal average value for esteem in each case.

When only a part of the group is motivated by prestige, these agents are not necessarily those that have the highest rank. Sometimes agents that are not motivated by prestige are taken into the elite by the reception of prestige gifts emitted by others: the others' motivation can directly influence an agent position.

Hence, even if one individual agent has autonomous choices, there is no independence concerning its belonging to the elite nor to the values of its own esteem or motivation.

For example, it is here possible to interpret the motivation change constant as the "value" that the society gives to prestige compared to sharing. If the idea of getting prestigious is not very important in the group, the agents will do their best to make a lot of sharing gifts but will not necessarily try regularly to do prestige gifts, which is what we find in the evolution of our societies.

5.2 An artificial reputation to base new kinds of organisations?

Here we represent an idea of reputation that is very different from the one that is usually used is multi-agent systems. The representation that the agents have of each other is not social and is just related to individual interactions. First, in most societies, the idea that the whole group is looking at the actions of each is absent [9]. That difference is solved in a multi-agent model, where agents still perform strategic interactions, but where the memory of defection is remembered by the whole group of agents and thus creates a social punishment [5]. In that case the reputation is a direct results of the individual actions: there is no interaction between the different concomitant actions in the group as we can see in our model. In the societies described where the agents lead strategic interactions, the fact that each action influences the others' representations is only a secondary aspect. Thus, when the agent has a reputation, its possibility of actions does not change: it can still choose to act with someone or refuse, and choose to cooperate or defect. Actually that type of model does not allow us to capture two very important element of reputation building in societies that have been identified by sociologists and ethnologists. First there are a huge variety of behaviours that build reputation and have no other meaning in a group, and they are central in any study of reputation building [1]. The second aspect is the evolution of the structure of the society due to the fact that the agents transform their own behaviour according to their situation in society [10].

In our society, the act of exchange is the only way that the agents have to communicate with the others. Based on the circulation of goods and the global values of the group, a common understanding is built. From that understanding, the agents acquire a status and an ability to act that is related to a hierarchical distribution of the gift. It is a strong assumption about how the agents can signal intentions to the others that we picture here: the gift has its place in a process where it delivers a message accessible to all. This is a very important result in the sense that it represents a successful representation of the ethnologists' point of view on gift, and that it actually enabled us to create some stable organised societies (depending on parameters). The only important assumption to be able to build that kind of system is to suppose that agents do not act for their interest but to secure the group and their own reputation. Just that wish is enough to create real dynamics: there is no such thing as an individual relation and yet the individual can define his or her own place in the group. What we get here is a complete and complex representation for the agent (which includes a representation of all the others and a personal representation allowing its actions), without basing it on a sophisticated individual cognition. The coherence of the evolution of reputation and of actions is kept, although only the structure of the exchange is what constraint the understanding. The societal element is predominant in the actions of the agents, but they still have completely individual choices, which is a good representation of the sociologist point of view on society.

Acknowledgements

Cirad supported the building of these models and the simulations that were led.

References

1. Aguilar J. L., 1984, Trust and Exchange: Expressive and Instrumental Dimensions of Reciprocity in a Peasant Community, Ethos, vol 12, n. 1, 3-29.
2. Boas, F., 1966, Kwakiutl Ethnography, The University of Chicago Press, Chicago.
3. Bousquet, F. and I. Bakam and H. Proton and C. Le Page (1998) Cormas: Common-Pool Resources and Multi-Agent Systems, Lecture Notes in Artificial Intelligence 1416.
4. Bousquet F., 1994, Des milieux, des poissons, des hommes : étude par simulations multi-agents, Orstom editions.
5. Chi Wong H., Sycara K., 1999, Adding Security and Trust to Multi-Agent Systems, In : Proceedings of « Deception, Fraud and Trust in Agent Societies », Cristiano Castelfranchi and Yao-Hua Tan and Rico Falcone and Babak Sadighi Firozabadi, pp 149-162, National Research Council, Rome, Italy.
6. Doran J., Palmer M., 1995, The EOS Project: Integrating two Models of paleolithic social change. In : Artificial Societies. The Computer Simulation of Social Life, Conte, R., Gilbert, N. (eds), UCL Press, London, pp 103-125.
7. Gilbert N., 1995, Emergence in Social Simulation, In : Artificial Societies. The Computer Simulation of Social Life, Conte, R., Gilbert, N. (eds), UCL Press, London, 144-156.
8. Godelier, M., 1996, L'énigme du don, Fayard, Paris.
9. Lahno B., 1995, Trust and Strategic Rationality, Rationality and society, vol 7, n. 4, pp 442-464.
10. Mauss M., 1968, L'énigme du don, In : Sociologie et anthropologie, PUF, Paris.
11. Nicolas G. 1986, Don rituel et échange marchand dans une société sahélienne, Institut d'ethnologie, Paris.
12. Rouchier J. et Bousquet F., 1999, Non-Merchant economy and multi-agent system: an analysis of structuring exchanges, In : Sichman J. S., Conte R. et Gilbert N. (eds), Multi-Agent Systems and Agent-Based Simulation, LNAI, vol 1534, Springer-Verlag, Berlin, 111-123.
13. Rouchier, J., Barreteau, O., Bousquet, F.: Evolution and Coevolution of Individuals and Groups. In: Yves Demazeau (ed), Proceedings of the Third International Conference on Multi-Agent Systems, IEEE Los Alamitos, USA (1998) 254-260.
14. Rouchier J., 1996, Le Potlatch : Une modélisation par les systèmes multi-agents, Master Thesis, Université d'Orléans.
15. Rouchier 2001, Apparition of reputation in an artificial society organised by a gift system, JASSS special issue.
16. Van de Ven Jeroen, 2000, The economics of the Gift, http://cwis.kub.nl/~few5/center/phdstud/ven/, non publié.
17. Weber J., 1985, C=R-I, My God, My Gold ! (Réflexion sur la Portée du concept de consumation, La revue du MAUSS, vol.13.